ЗЕРТХАНАЛЫҚ ЖҰМЫСТЫ орындауға арналған
ӘДІСТЕМЕЛІК НҰСҚАУЛЫҚ

Huawei eNSP симуляторын қолданған жағдайда

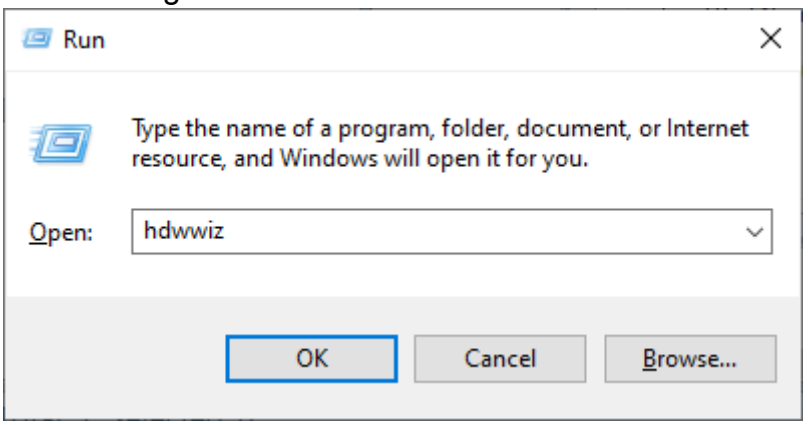
Тақырыбы: Network Automation. Python
Жұмыстың мақсаты: HCIA-Datacom деңгейін меңгеру

Жұмыстың орындалу қадамы:

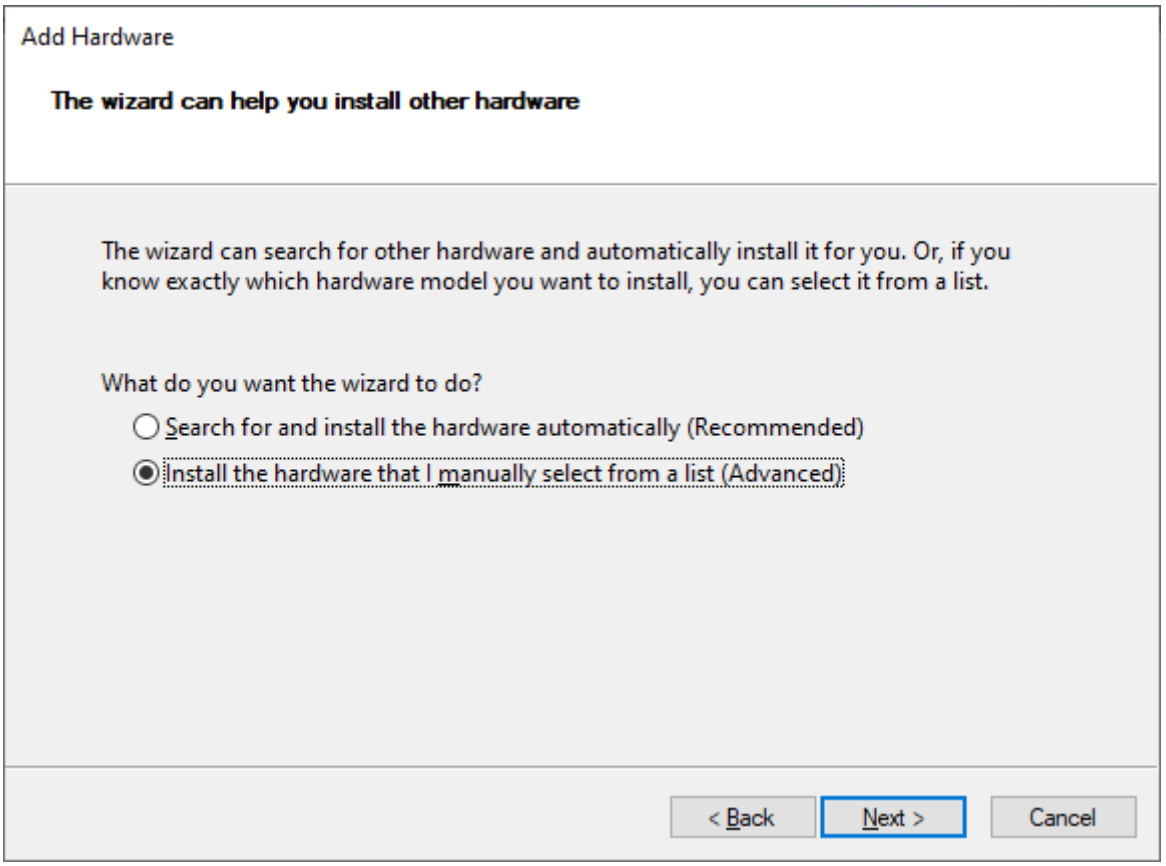
1) Windows операциялық жүйесінде Loopback interface құру
2) Windows операциялық жүйесімен eNSP симуляторын байланыстыру
3) Ubuntu Linux дистрибутивімен eNSP симуляторын байланыстыру
4) Telnet конфигурациялау
5) Python TELNET Library
6) SSH конфигурациялау
7) Python Netmiko Library
8) Python NAPALM Library

1-қадам: Windows операциялық жүйесінде Loopback interface құру

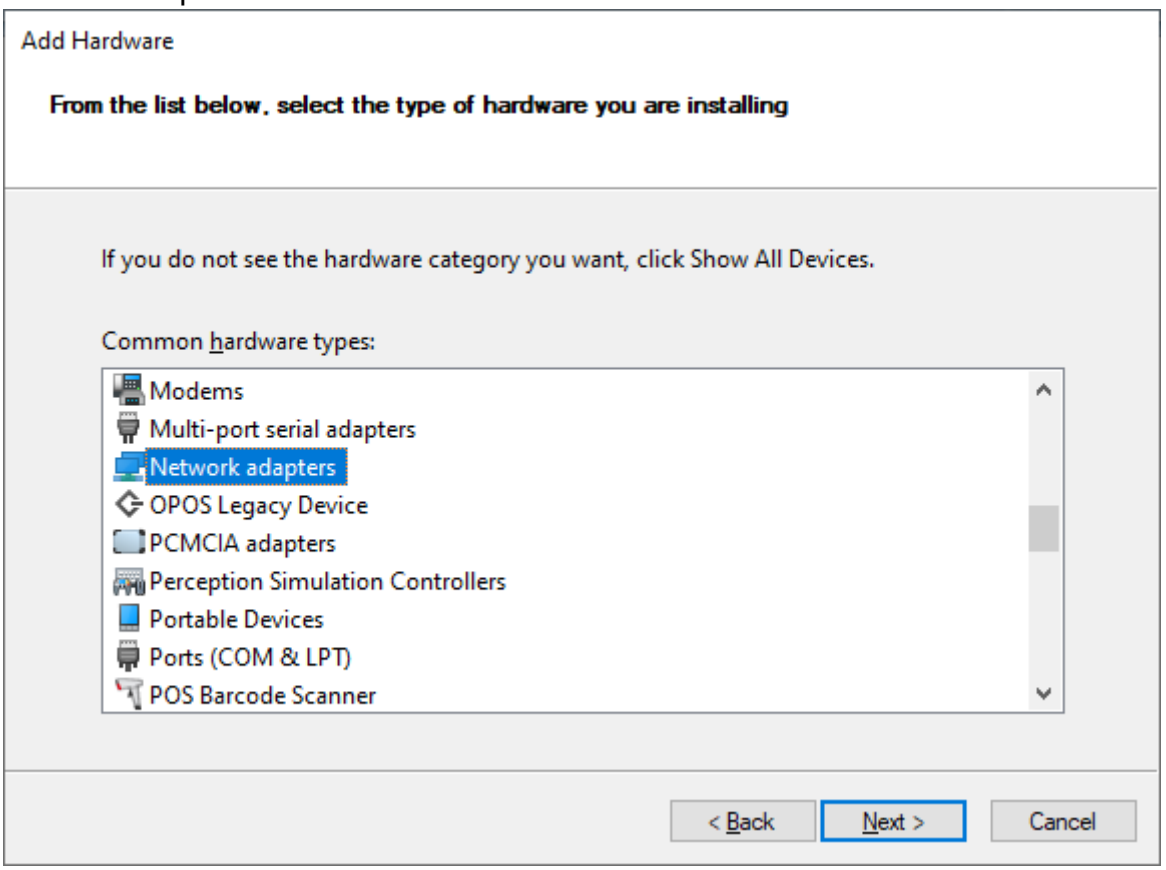
Windows logo +R



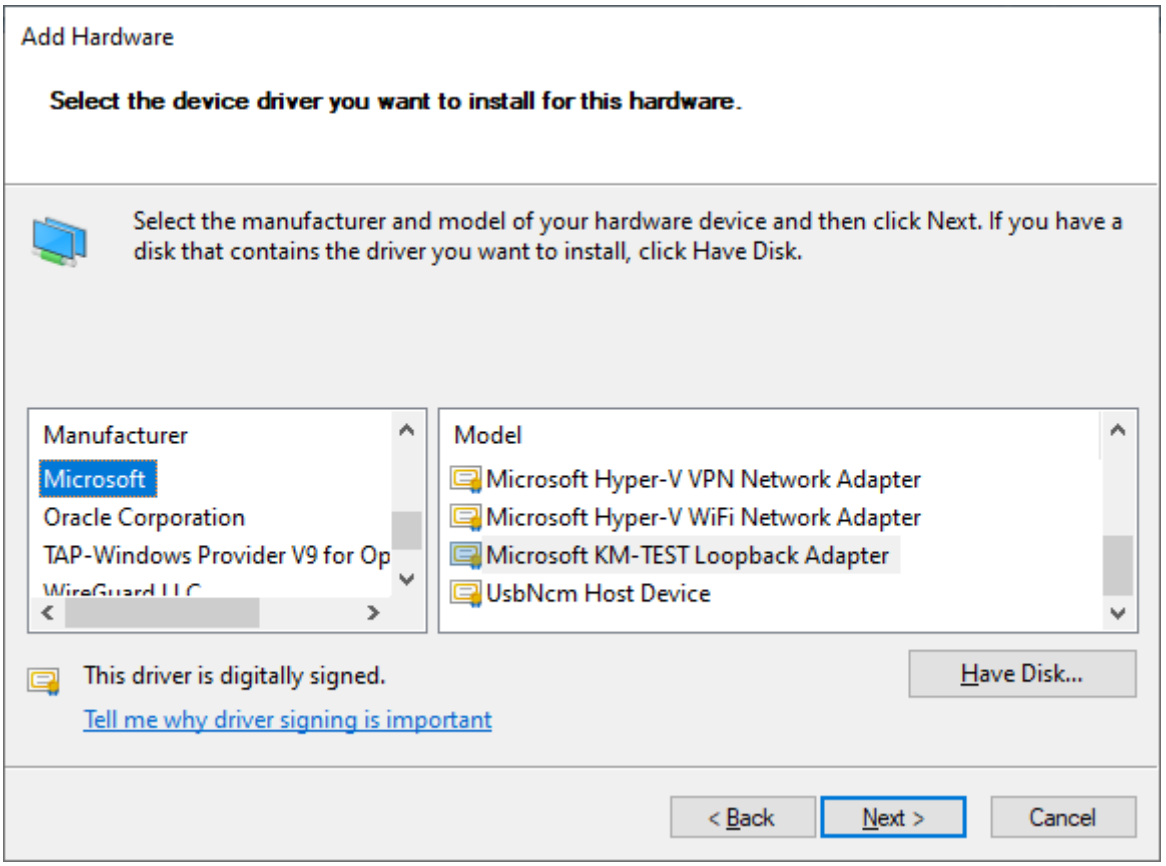
install the hardware ...



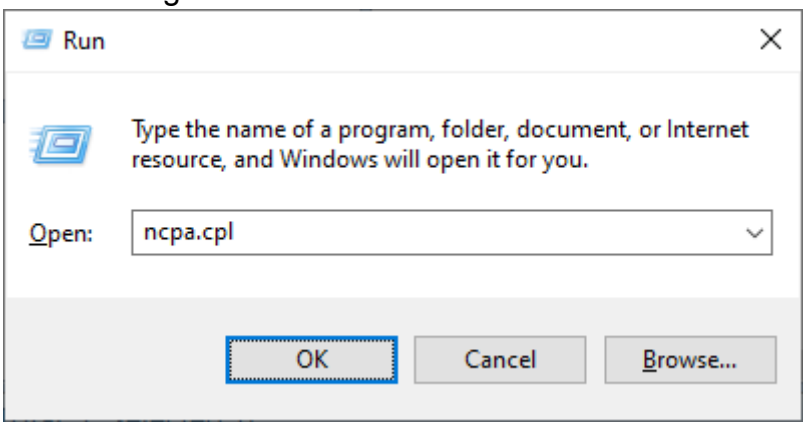
Network adapters



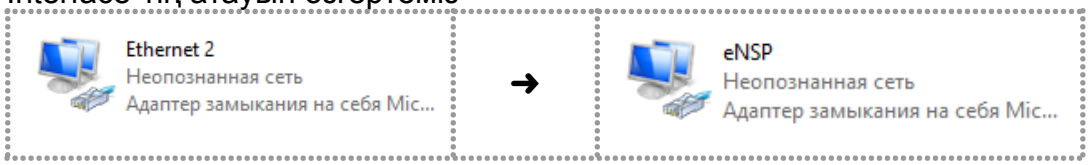
Microsoft ➜ Microsoft KM-TEST Loopback Adapter



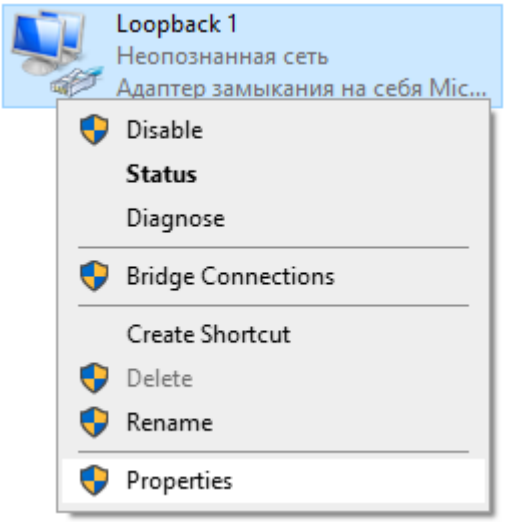
Windows logo +R

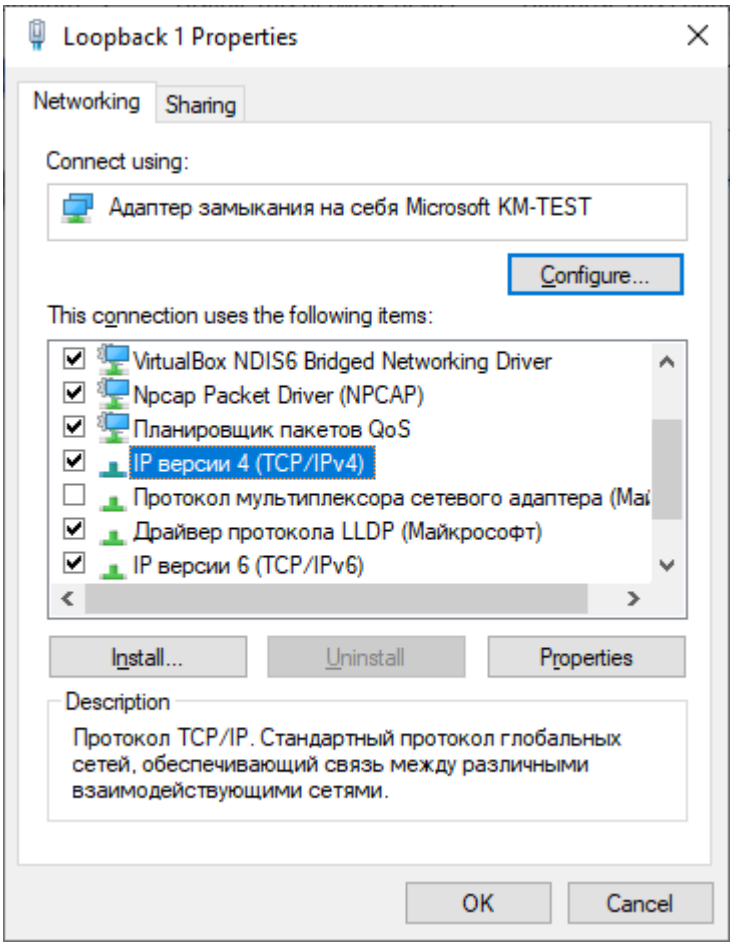


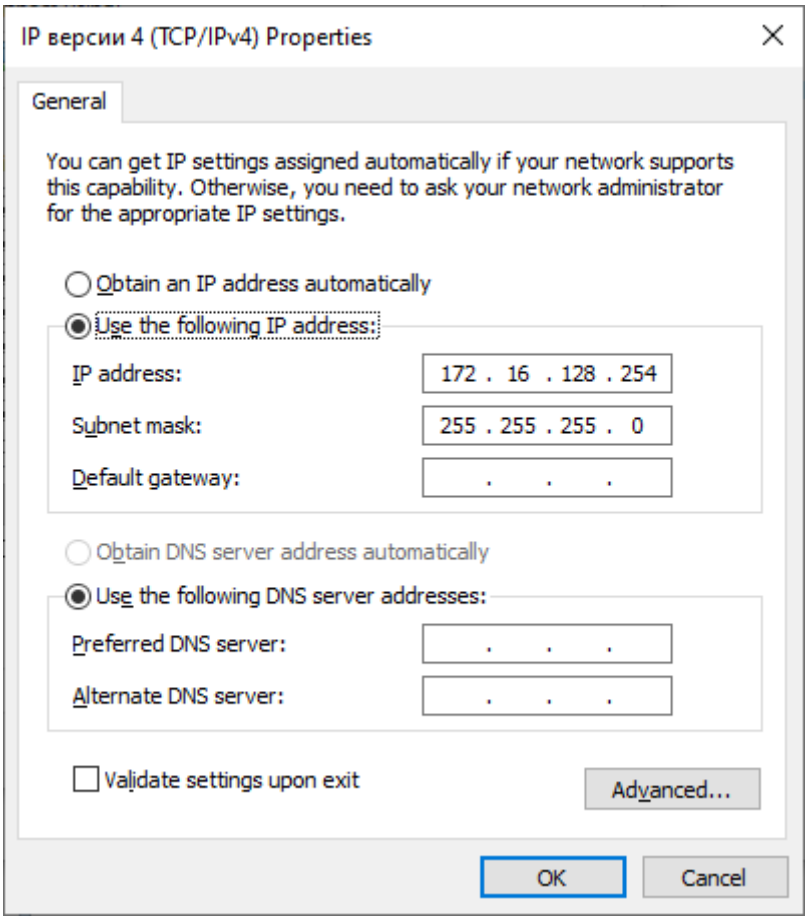
Interface-тің атауын өзгертеміз



Loopback interface ➜ Properties ➜ TCP/IPv4 ➜ IP address енгізу

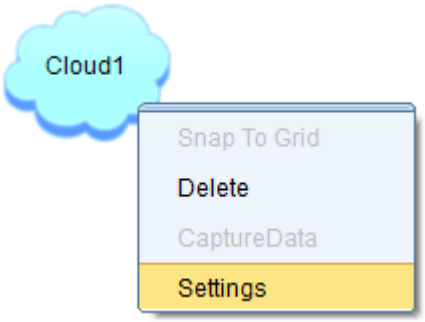

Loopback 1
Неопознанная сеть
Адаптер замыкания на себя Mic...
Disable
Status
Diagnose
Bridge Connections
Create Shortcut
Delete
Rename
Properties


Loopback 1 Properties
Networking
Sharing
Connect using:
Адаптер замыкания на себя Microsoft KM-TEST
Configure...
This connection uses the following items:
VirtualBox NDIS6 Bridged Networking Driver
Npcap Packet Driver (NPCAP)
Планировщик пакетов QoS
IP версии 4 (TCP/IPv4)
Протокол мультиплексора сетевого адаптера (Mai
Драйвер протокола LLDP (Майкрософт)
IP версии 6 (TCP/IPv6)
Install...
Uninstall
Properties
Description
Протокол TCP/IP. Стандартный протокол глобальных сетей, обеспечивающий связь между различными взаимодействующими сетями.
OK
Cancel

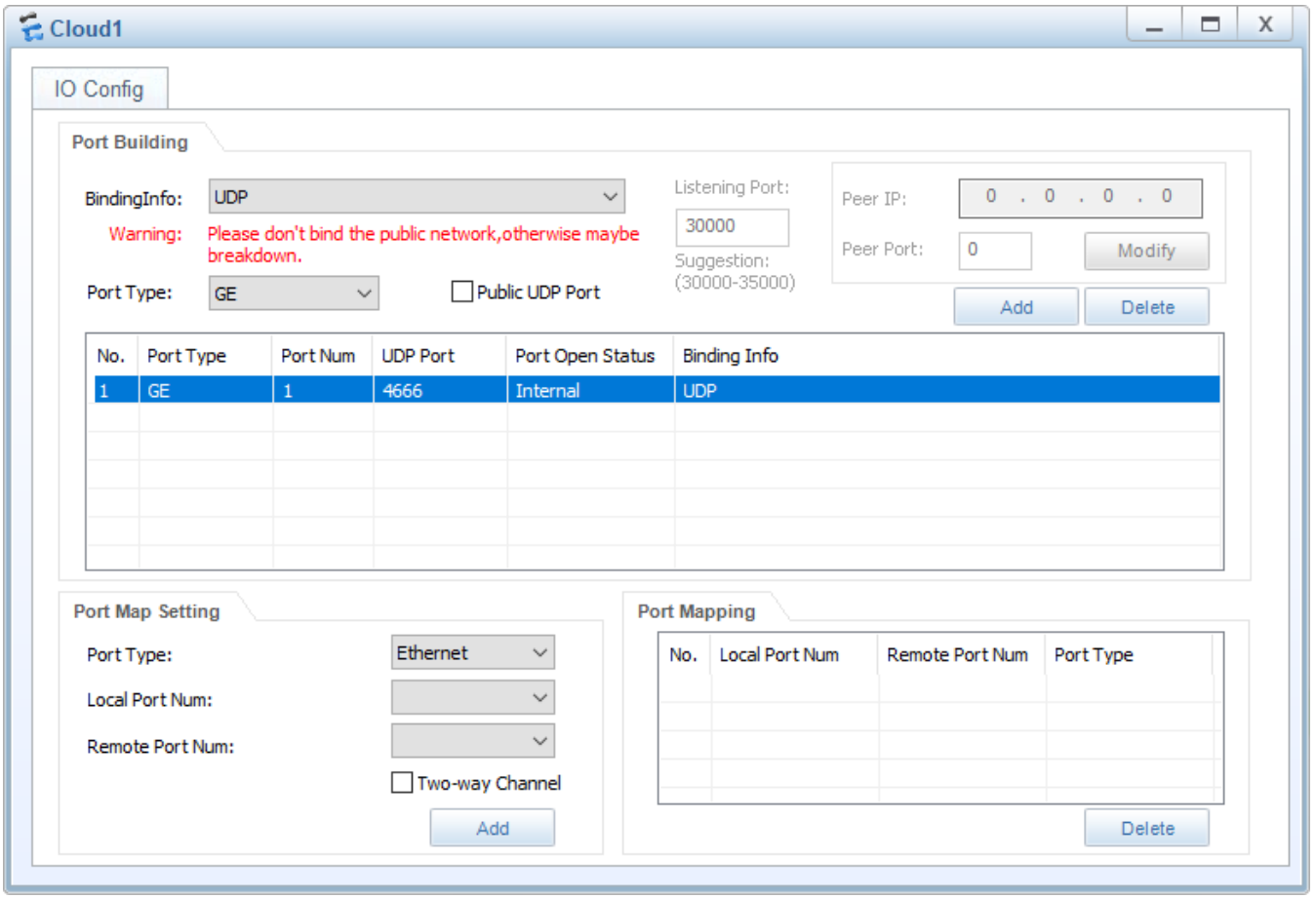


PC/Laptop ➜ Reboot

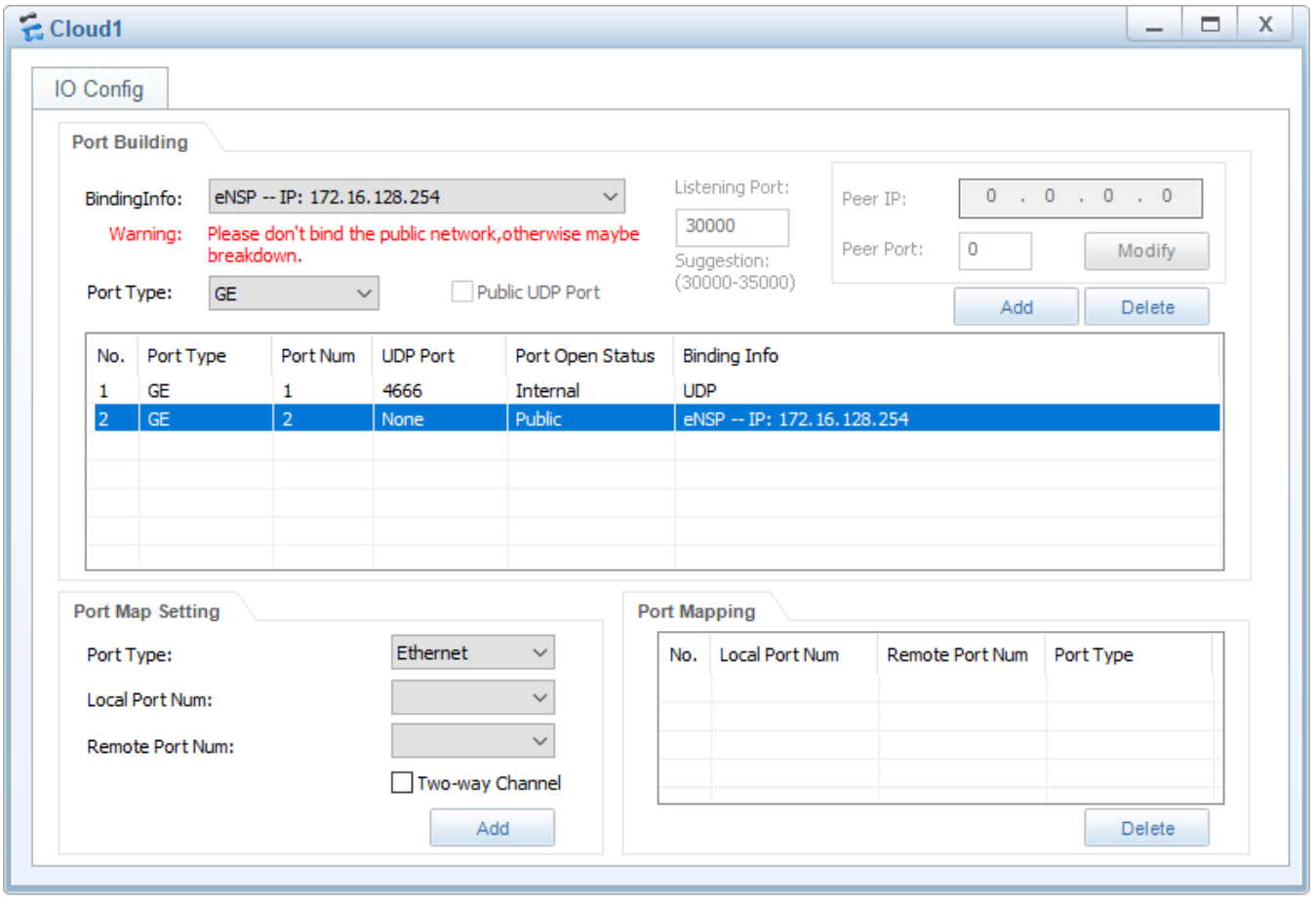
2-қадам: Windows операциялық жүйесімен eNSP симуляторын байланыстыру



Port Building ➜ BindingInfo: UDP
➜ Port Type: GE
➜ “Add” батырманы басу



Port Building:
➜ BindingInfo: eNSP
➜ Port Type: GE
➜ “Add” батырманы басу



Port Map Setting:

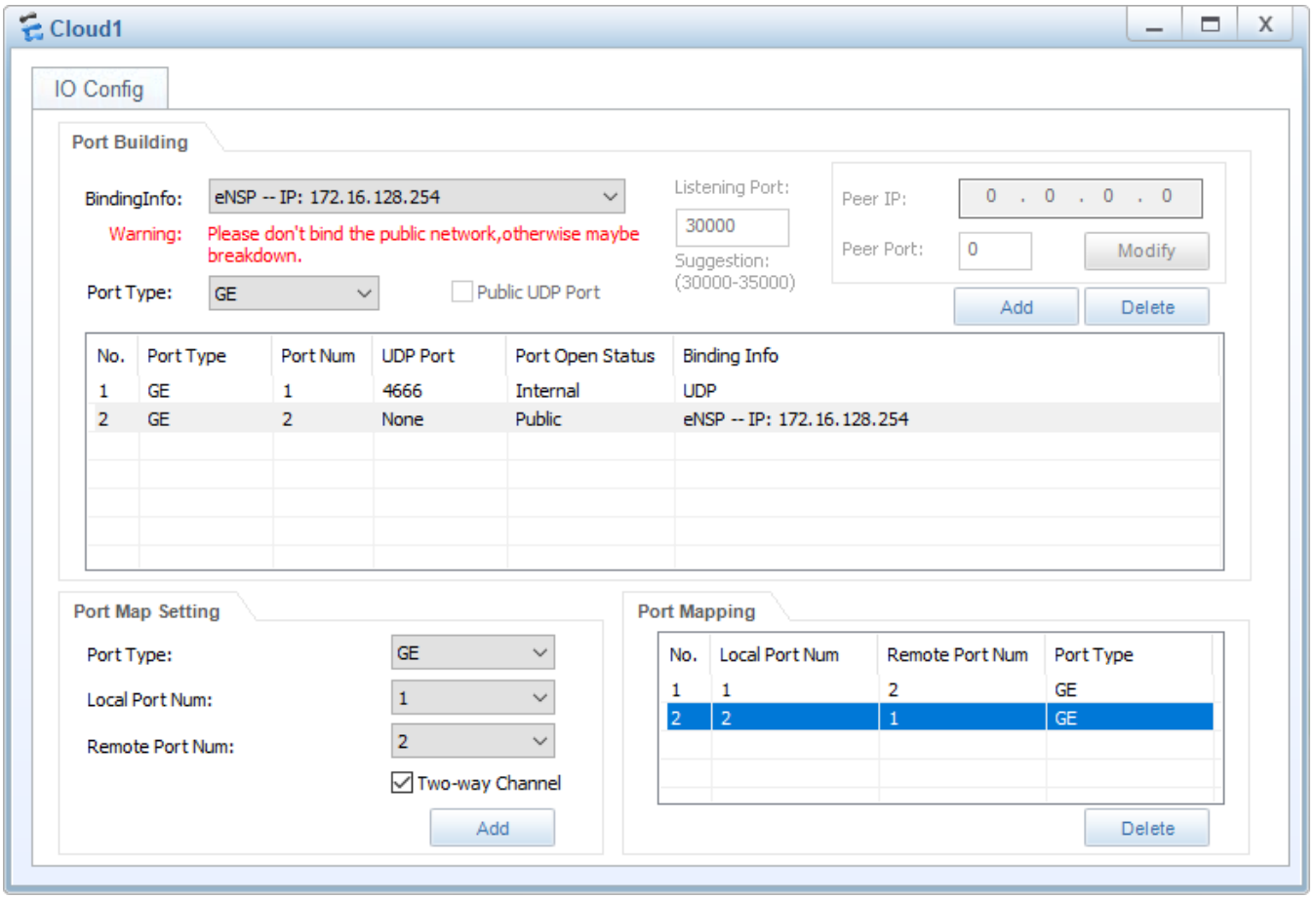
➜ Port Type: GE

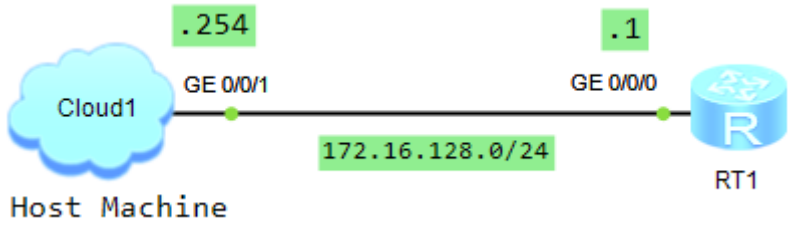
➜ Local Port Num: 1

➜ Remote Port Num: 2

☑ Two-way Channel

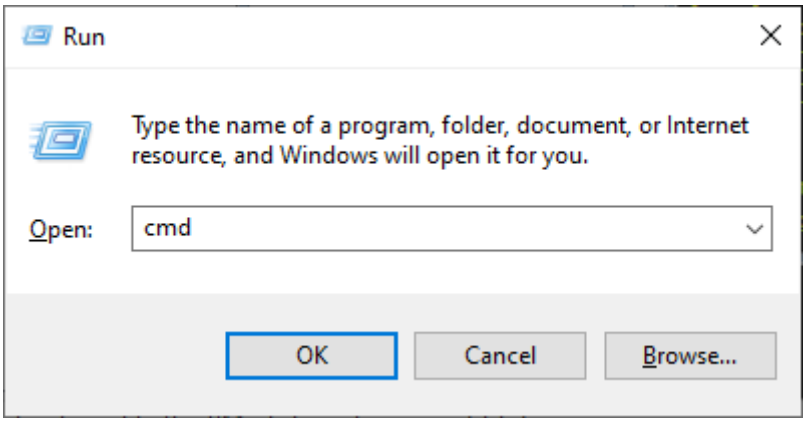
➜ “Add” батырманы басу

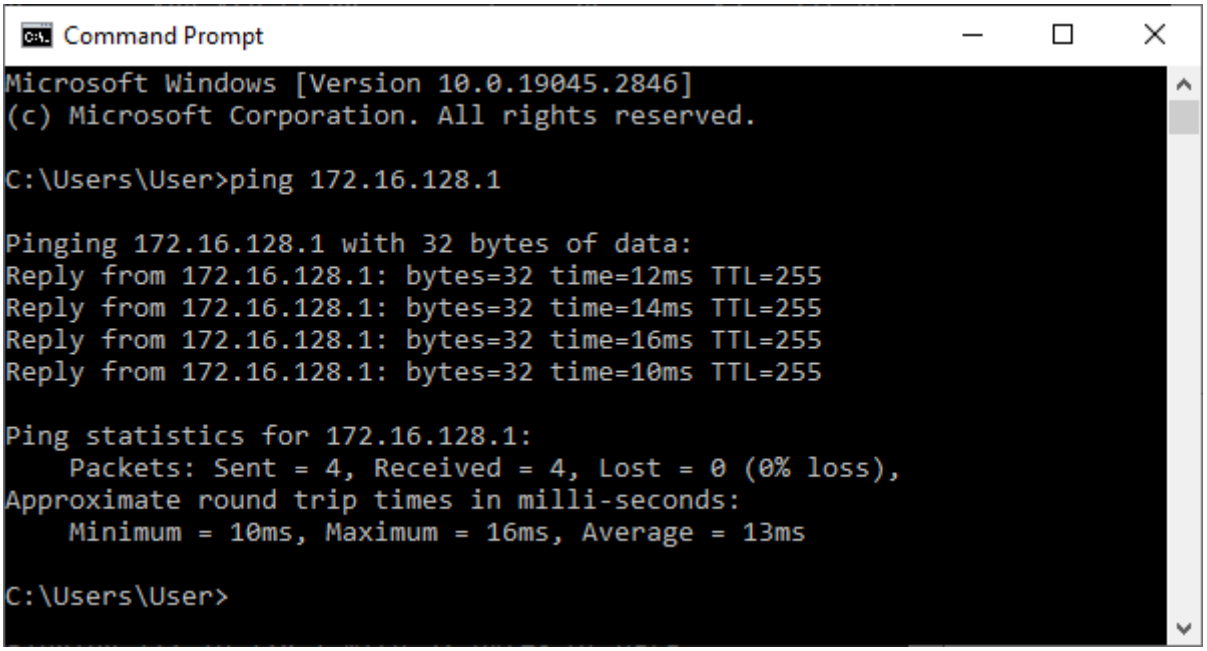





```
<Huawei> undo terminal logging
<Huawei> system-view
[Huawei] sysname RT1

[RT1] interface g0/0/0
[RT1-GigabitEthernet0/0/0] ip address 172.16.128.1 24

[RT1] display ip int brief
Interface                         IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0              172.16.128.1/24      up         up
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
NULL0                             unassigned           up         up(s)
```

Windows logo +R





3-қадам: Ubuntu Linux дистрибутивімен eNSP симуляторын байланыстыру

Download Linux images (OVF Files)
https://drive.google.com/drive/folders/1cUMzXEw4BFvp3dGhMJQiGjQnVS0b3MTB?usp=sharing
Download VMware Workstation Pro
https://drive.google.com/drive/folders/1xPeOKfdeOzGdEHJRhYktJThgL6-xjkHy?usp=sharing

Link YouTube Channel
https://www.youtube.com/playlist?list=PLqzHlBIxhTu2wlO2GhGKEV5edlE0z7SSb

...

login: student
Password: 123

student@ubuntu22-04~$ `ip address`

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 00:0c:29:a0:a0:d2 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.0.104/24 metric 100 brd 192.168.0.255 scope global dynamic
       valid_lft 7171sec preferred_lft 7171sec
    inet6 fe80::20c:29ff:fea0:a0d2/64 scope link
       valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default
    link/ether 00:0c:29:a0:a0:dc brd ff:ff:ff:ff:ff:ff
    altname enp2s5
```

student@ubuntu22-04~$ `sudo nano /etc/netplan/00-installer-config.yaml`

```
network:
  ethernets:
    ens33:
      dhcp4: true
    ens37:
      addresses: [172.16.128.5/24]
  version: 2
```

Ctrl+O ➜ Enter
Ctrl+X

student@ubuntu22-04~$ `sudo netplan apply`

student@ubuntu22-04~$ `ip address`

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 00:0c:29:a0:a0:d2 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.0.104/24 metric 100 brd 192.168.0.255 scope global dynamic
       valid_lft 5393sec preferred_lft 5393sec
    inet6 fe80::20c:29ff:fea0:a0d2/64 scope link
       valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 00:0c:29:a0:a0:dc brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 172.16.128.5/24 brd 172.16.128.255 scope global ens37
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fea0:a0dc/64 scope link
       valid_lft forever preferred_lft forever
```

student@ubuntu22-04~$ `ping -c4 172.16.128.254`

```
64 bytes from 172.16.128.254: icmp_seq=1 ttl=128 time=0.453 ms
64 bytes from 172.16.128.254: icmp_seq=2 ttl=128 time=0.803 ms
64 bytes from 172.16.128.254: icmp_seq=3 ttl=128 time=0.466 ms
64 bytes from 172.16.128.254: icmp_seq=4 ttl=128 time=0.494 ms
```

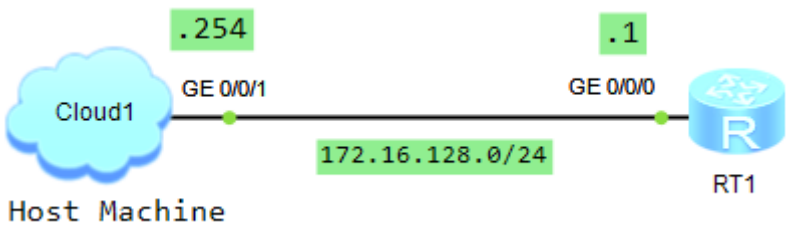
student@ubuntu22-04~$ `ping -c4 172.16.128.1`

```
64 bytes from 172.16.128.1: icmp_seq=1 ttl=255 time=11.1 ms
64 bytes from 172.16.128.1: icmp_seq=2 ttl=255 time=7.69 ms
64 bytes from 172.16.128.1: icmp_seq=3 ttl=255 time=13.0 ms
64 bytes from 172.16.128.1: icmp_seq=4 ttl=255 time=9.76 ms
```

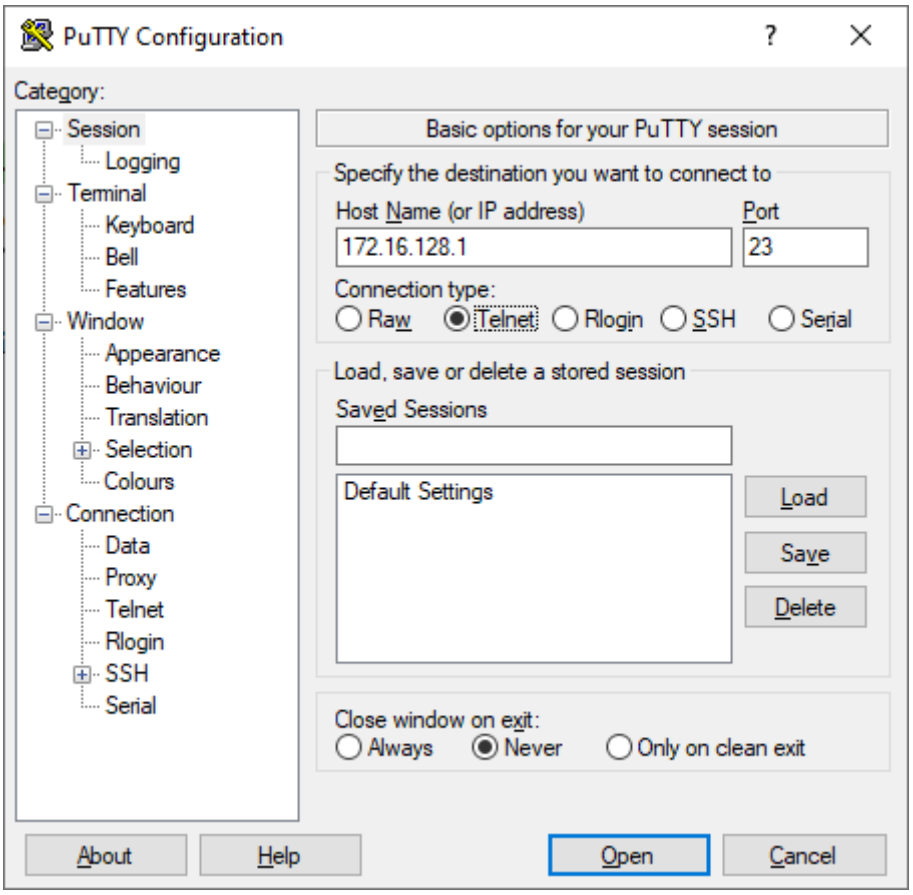
4-қадам: Telnet конфигурациялау (Configure Telnet Remote Management)

```
[RT1] display telnet server status
TELNET IPV4 server                    :Enable
TELNET IPV6 server                    :Enable
TELNET server port                    :23
```

```
[RT1] user-interface vty 0 4
[RT1-ui-vty0-4] authentication-mode aaa

[RT1] aaa
[RT1-aaa] local-user user1 password cipher user@123
[RT1-aaa] local-user user1 privilege level 15
[RT1-aaa] local-user user1 service-type telnet
```



Download PuTTY
https://putty.org/



```
Login authentication


Username:user1
Password:
<RT1>system-view
Enter system view, return user view with Ctrl+Z.
[RT1]
```

student@ubuntu22-04~$ `telnet`

```
telnet> open 172.16.128.1
Trying 172.16.128.1...
Connected to 172.16.128.1.
Escape character is '^]'.


Login authentication


Username:user1
Password:
<RT1>system-view
Enter system view, return user view with Ctrl+Z.
[RT1]
```

5-қадам: Python TELNET Library

student@ubuntu22-04~$ `python3 --version`

```
Python 3.10.6
```

student@ubuntu22-04~$ `ls -l`

```
-rw-rw-r-- 1 student student 0 Apr 23 04:21 script1_telnet.py
-rw-rw-r-- 1 student student 0 Apr 23 04:21 script2_telnet.py
-rw-rw-r-- 1 student student 0 Apr 23 04:22 script3_telnet.py
```

student@ubuntu22-04~$ `nano script1_telnet.py`

```
import telnetlib

host = '172.16.128.1'
user = 'user1'
password = 'user@123'

tn = telnetlib.Telnet(host)
```

```
tn.read_until(b'Username:')
tn.write(user.encode('ascii') + b'\n')
tn.read_until(b'Password:')
tn.write(password.encode('ascii') + b'\n')

print('Connection to ' + '172.16.128.1' + ' is Successful')

tn.close()
```

Ctrl+O ➜ Enter
Ctrl+X

student@ubuntu22-04~$ `python3 script1_telnet.py`

```
Connection to 172.16.128.1 is Successful
```

student@ubuntu22-04~$ `nano script2_telnet.py`

```
import telnetlib
import time

host = '172.16.128.1'
user = 'user1'
password = 'user@123'
UserPrompt = '>'
ConfigPrompt = ']'

tn = telnetlib.Telnet(host)

tn.read_until(b'Username:')
tn.write(user.encode('ascii') + b'\n')
tn.read_until(b'Password:')
tn.write(password.encode('ascii') + b'\n')

print('Connection to ' + '172.16.128.1' + ' is Successful')

UserMode = tn.read_until(UserPrompt.encode('ascii'))
print(UserMode.decode('ascii'))

tn.write(b'system-view \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.close()
```

Ctrl+O ➜ Enter
Ctrl+X

student@ubuntu22-04~$ `python3 script2_telnet.py`

```
Connection to 172.16.128.1 is Successful

<RT1>
system-view
Enter system view, return user view with Ctrl+Z.
[RT1]
student@ubuntu22-04:~$
```

student@ubuntu22-04~$ `nano script3_telnet.py`

```
import telnetlib
import time

host = '172.16.128.1'
user = 'user1'
password = 'user@123'
```

```
UserPrompt = '>'
ConfigPrompt = ']'

tn = telnetlib.Telnet(host)

tn.read_until(b'Username:')
tn.write(user.encode('ascii') + b'\n')
tn.read_until(b'Password:')
tn.write(password.encode('ascii') + b'\n')

print('Connection to ' + '172.16.128.1' + ' is Successful')

UserMode = tn.read_until(UserPrompt.encode('ascii'))
print(UserMode.decode('ascii'))

tn.write(b'system-view \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.write(b'interface LoopBack 0 \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.write(b'ip address 50.0.1.1 24 \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.write(b'display ip interface brief \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.close()
```

Ctrl+O ➜ Enter
Ctrl+X

student@ubuntu22-04~$ `python3 script3_telnet.py`

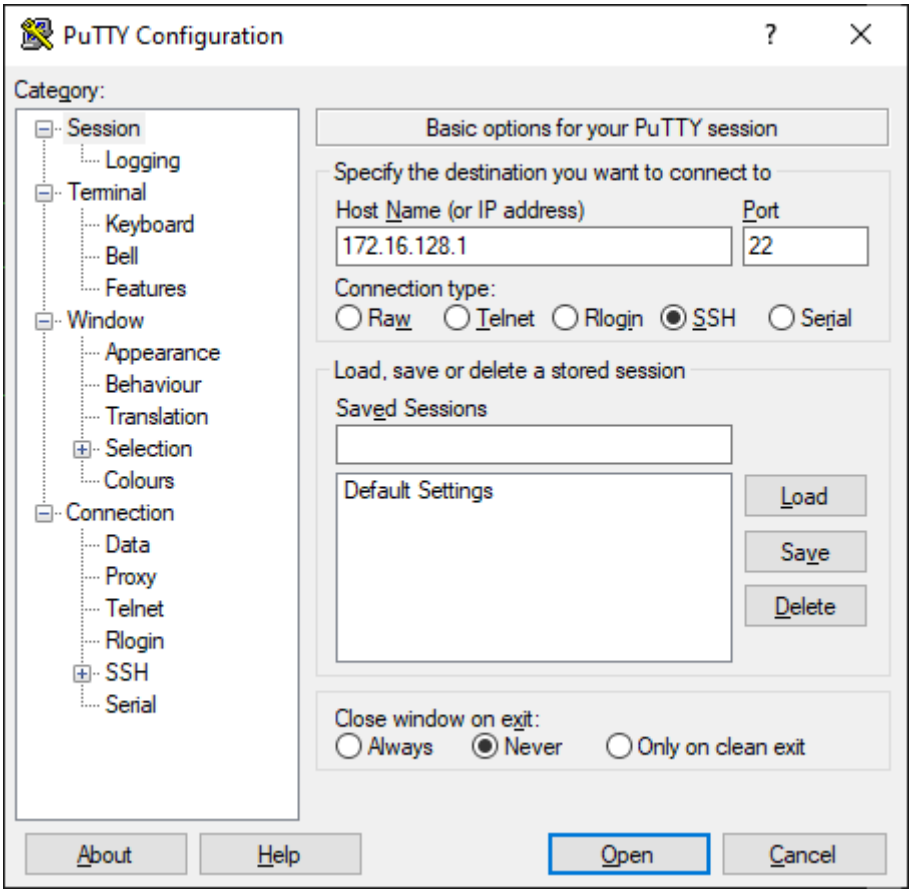
```
Connection to 172.16.128.1 is Successful

<RT1>
system-view
Enter system view, return user view with Ctrl+Z.
[RT1]
interface LoopBack 0
[RT1-LoopBack0]
ip address 50.0.1.1 24
[RT1-LoopBack0]
display ip interface brief

Interface                         IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0              172.16.128.1/24      up         up
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
LoopBack0                         50.0.1.1/24          up         up(s)
NULL0                             unassigned           up         up(s)
[RT1-LoopBack0]
student@ubuntu22-04:~$
```

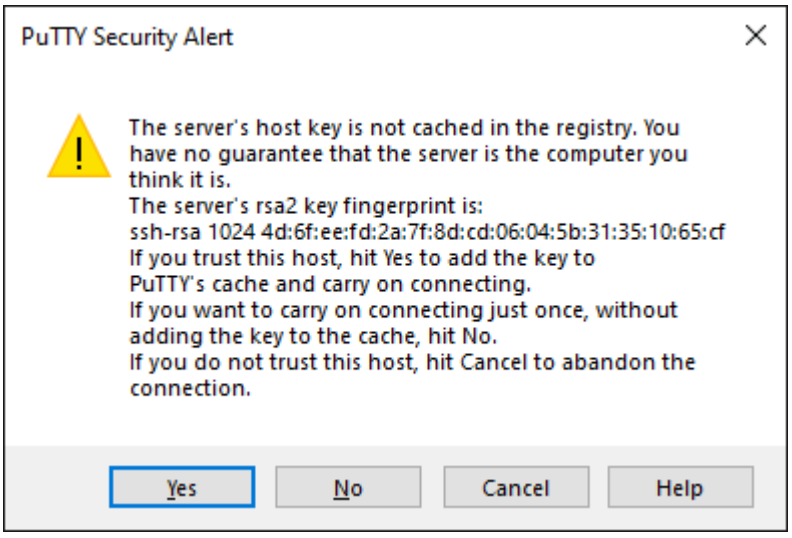
6-қадам: SSH конфигурациялау (Configure SSH Remote Management)

```
[RT1] rsa local-key-pair create
```

```
Confirm to replace them? (y/n)[n]: y
Input the bits in the modulus[default = 512]: 2048

[RT1] user-interface vty 0 4
[RT1-ui-vty0-4] authentication-mode aaa
[RT1-ui-vty0-4] protocol inbound all
[RT1-ui-vty0-4] display this

[RT1] aaa
[RT1-aaa] local-user user2 password cipher user@123
[RT1-aaa] local-user user2 privilege level 15
[RT1-aaa] local-user user2 service-type ssh

[RT1] stelnet server enable
Succeeded in starting the STELNET server
```

Remote Access over PuTTY (Windows)






```
login as: user2
user2@172.16.128.1's password:

<RT1>system-view
Enter system view, return user view with Ctrl+Z.
[RT1]
```

Remote Access over Terminal (Ubuntu Linux)

<RT1> `display ssh server status`

```
SSH version                          :1.99
SSH connection timeout               :60 seconds
SSH server key generating interval   :0 hours
SSH Authentication retries           :3 times
SFTP Server                          :Disable
Stelnet server                       :Enable
```

student@ubuntu22-04~$ `systemctl status ssh`

```
Active: active (running)
```

student@ubuntu22-04~$ `sudo nano .ssh/config`

```
Ciphers aes128-ctr,aes192-ctr,aes256-ctr,aes128-cbc,3des-cbc
KexAlgorithms +diffie-hellman-group-exchange-sha1,diffie-hellman-group1-sha1
HostKeyAlgorithms=+ssh-rsa
```

Ctrl+O ➜ Enter
Ctrl+X

student@ubuntu22-04~$ `ssh user2@172.16.128.1`

```
The authenticity of host '172.16.128.1 (172.16.128.1)' can't be established.
RSA key fingerprint is SHA256:RP/9qVLMXrexBrzTCrnOTKtxFuh36ZOjo/6NdUhy/D8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.16.128.1' (RSA) to the list of known hosts.
user2@172.16.128.1's password:

<RT1>system-view
Enter system view, return user view with Ctrl+Z.
[RT1]
```

7-қадам: Python NetMiko Library

Paramiko – SSH2 protocol library
https://pypi.org/project/paramiko/

Netmiko – Multi-vendor library to simplify CLI connections to network devices
https://pypi.org/project/netmiko/

student@ubuntu22-04~$ `ping -c4 google.com`

```
64 bytes from 216.58.210.142: icmp_seq=1 ttl=111 time=69.0 ms
64 bytes from 216.58.210.142: icmp_seq=2 ttl=111 time=69.0 ms
64 bytes from 216.58.210.142: icmp_seq=3 ttl=111 time=69.2 ms
64 bytes from 216.58.210.142: icmp_seq=4 ttl=111 time=68.9 ms
```

student@ubuntu22-04~$ `sudo apt update`
student@ubuntu22-04~$ `sudo apt install -y build-essential libssl-dev libffi-dev`

student@ubuntu22-04~$ `python3 --version`

```
Python 3.10.6
```

student@ubuntu22-04~$ `sudo apt install -y python3-pip`
student@ubuntu22-04~$ `pip3 --version`

```
pip 22.0.2 from /usr/lib/python3/dist-packages/pip (python 3.10)
```

student@ubuntu22-04~$ `python3 -m venv netmiko_vrp`
student@ubuntu22-04~$ `source netmiko_vrp/bin/activate`
(netmiko_vrp) student@ubuntu22-04~$ `deactivate`

student@ubuntu22-04~$ `source netmiko_vrp/bin/activate`
(netmiko_vrp) student@ubuntu22-04~$ `python -m pip install paramiko`
(netmiko_vrp) student@ubuntu22-04~$ `python -m pip install netmiko`
(netmiko_vrp) student@ubuntu22-04~$ `python -m pip list`

```
Package        Version
-------------  -------
bcrypt         4.0.1
cffi           1.15.1
cryptography   40.0.2
future         0.18.3
netmiko        4.1.2
ntc-templates  3.3.0
paramiko       3.1.0
pip            22.0.2
pycparser      2.21
PyNaCl         1.5.0
pyserial       3.5
PyYAML         6.0
scp            0.14.5
setuptools     59.6.0
six            1.16.0
tenacity       8.2.2
textfsm        1.1.2
```

student@ubuntu22-04~$ `ls -l`

```
-rw-rw-r-- 1 student student 276 Apr 23 04:24 script1_netmiko.py
```

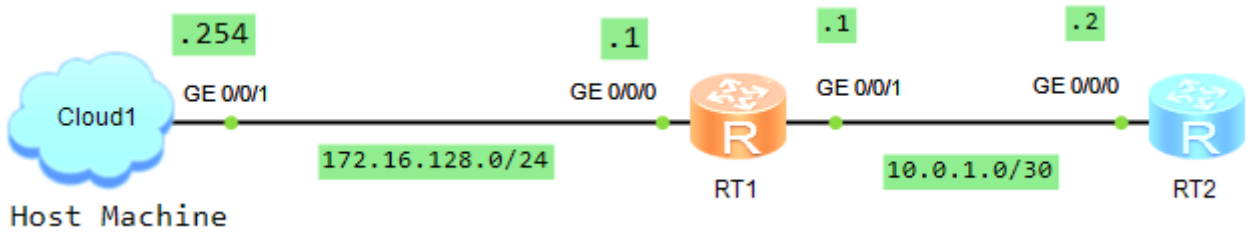
student@ubuntu22-04~$ `nano script1_netmiko.py`

```
from netmiko import ConnectHandler

AR2220 = {
      'device_type': 'huawei',
      'host': '172.16.128.1',
      'username': 'user2',
      'password': 'user@123'
}

net_connect = ConnectHandler(**AR2220)
output = net_connect.send_command('display ip interface brief')
print(output)
```

Ctrl+O ➜ Enter
Ctrl+X

(netmiko_vrp) student@ubuntu22-04~$ `python script1_netmiko.py`

```
Interface                     IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0          172.16.128.1/24      up         up
GigabitEthernet0/0/1          unassigned           down       down
GigabitEthernet0/0/2          unassigned           down       down
LoopBack0                     50.0.1.1/24          up         up(s)
NULL0                         unassigned           up         up(s)
```



student@ubuntu22-04~$ `nano script2_netmiko.py`

```
from netmiko import ConnectHandler

AR2220 = {
      'device_type': 'huawei',
      'host': '172.16.128.1',
      'username': 'user2',
      'password': 'user@123'
}

net_connect = ConnectHandler(**AR2220)
output = net_connect.send_command('display current-configuration interface g0/0/1')
print(output)

commands = ['interface GigabitEthernet 0/0/1', 'ip address 10.0.1.1 30', 'display
this', 'ospf 1 router-id 1.1.1.1', 'area 0', 'network 10.0.1.0 0.0.0.3', network
172.16.128.0 0.0.0.255']
output = net_connect.send_config_set(commands)
print(output)

output = net_connect.send_command('display current-configuration section ospf')
print(output)
```

Ctrl+O ➜ Enter
Ctrl+X

(netmiko_vrp) student@ubuntu22-04~$ `python script2_netmiko.py`

```
#
interface GigabitEthernet0/0/1
#
return
system-view
Enter system view, return user view with Ctrl+Z.
[RT1]interface GigabitEthernet 0/0/1
[RT1-GigabitEthernet0/0/1]ip address 10.0.1.1 30
[RT1-GigabitEthernet0/0/1]display this
[V200R003C00]
#
interface GigabitEthernet0/0/1
 ip address 10.0.1.1 255.255.255.252
#
return
[RT1-GigabitEthernet0/0/1]ospf 1 router-id 1.1.1.1
[RT1-ospf-1]area 0
[RT1-ospf-1-area-0.0.0.0]network 10.0.1.0 0.0.0.3
[RT1-ospf-1-area-0.0.0.0]network 172.16.128.0 0.0.0.255
[RT1-ospf-1-area-0.0.0.0]return
<RT1>
[V200R003C00]
#
ospf 1 router-id 1.1.1.1
 area 0.0.0.0
  network 10.0.1.0 0.0.0.3
  network 172.16.128.0 0.0.0.255
#
return
```

8-қадам: Python NAPALM Library

NAPALM's documentation
https://napalm.readthedocs.io

NAPALM (Network Automation and Programmability Abstraction Layer with Multivendor support) is a Python library that implements a set of functions to interact with different network device Operating Systems using a unified API.

NAPALM Huawei VRP
https://github.com/napalm-automation-community/napalm-huawei-vrp
https://pypi.org/project/napalm-huawei-vrp/

student@ubuntu22-04~$ `python3 -m venv napalm_vrp`
student@ubuntu22-04~$ `source napalm_vrp/bin/activate`
(napalm_vrp) student@ubuntu22-04~$ `python -m pip install napalm-huawei-vrp`
(napalm_vrp) student@ubuntu22-04~$ `python -m pip list`

student@ubuntu22-04~$ `ls -l`

```
-rw-rw-r-- 1 student student 283 Apr 23 04:50 script1_napalm.py
-rw-rw-r-- 1 student student 294 Apr 23 04:53 script2_napalm.py
-rw-rw-r-- 1 student student 278 Apr 23 04:55 script3_napalm.py
-rw-rw-r-- 1 student student 288 Apr 23 04:56 script4_napalm.py
```

student@ubuntu22-04~$ `nano script1_napalm.py`

```
from napalm import get_network_driver
import pprint

driver = get_network_driver('huawei_vrp')
device = driver(hostname='172.16.128.1', username='user2', password='user@123')
device.open()

# Get Facts API. Return general device information
get_facts = device.get_facts()
pprint.pprint(get_facts)
```

Ctrl+O ➜ Enter
Ctrl+X

(napalm_vrp) student@ubuntu22-04~$ `python script1_napalm.py`

```
{'fqdn': 'Unknown',
 'hostname': 'RT1',
 'interface_list': ['GigabitEthernet0/0/0',
                    'GigabitEthernet0/0/1',
                    'GigabitEthernet0/0/2',
                    'NULL0'],
 'model': 'Unknown',
 'os_version': 'Unknown',
 'serial_number': [],
 'uptime': -1,
 'vendor': 'Huawei'}
```

student@ubuntu22-04~$ `nano script2_napalm.py`

```
from napalm import get_network_driver
import pprint
```

```
driver = get_network_driver('huawei_vrp')
device = driver(hostname='172.16.128.1', username='user2', password='user@123')
device.open()

# Get Interfaces IP API
get_interfaces_ip = device.get_interfaces_ip()
pprint.pprint(get_interfaces_ip)
```

Ctrl+O ➜ Enter
Ctrl+X

(napalm_vrp) student@ubuntu22-04~$ `python script2_napalm.py`

```
{'GigabitEthernet0/0/0': {'ipv4': {'172.16.128.1': {'prefix_length': 24}}}}
```

student@ubuntu22-04~$ `nano script3_napalm.py`

```
from napalm import get_network_driver
import pprint

driver = get_network_driver('huawei_vrp')
device = driver(hostname='172.16.128.1', username='user2', password='user@123')
device.open()

# Ping API
ping_remote_ip = device.ping('172.16.128.254')
pprint.pprint(ping_remote_ip)
```

Ctrl+O ➜ Enter
Ctrl+X

student@ubuntu22-04~$ `python3 script3_napalm.py`

```
{'success': {'packet_loss': 0,
             'probes_sent': 5,
             'results': [{'ip_address': '172.16.128.254', 'rtt': 10.0},
                         {'ip_address': '172.16.128.254', 'rtt': 10.0},
                         {'ip_address': '172.16.128.254', 'rtt': 10.0},
                         {'ip_address': '172.16.128.254', 'rtt': 1.0},
                         {'ip_address': '172.16.128.254', 'rtt': 10.0}],
             'rtt_avg': 8.0,
             'rtt_max': 10.0,
             'rtt_min': 1.0,
             'rtt_stddev': 0.0}}
```

student@ubuntu22-04~$ `nano script4_napalm.py`

```
from napalm import get_network_driver
import pprint

driver = get_network_driver('huawei_vrp')
device = driver(hostname='172.16.128.1', username='user2', password='user@123')
device.open()

# Send Any CLI command
send_command = device.cli(['display version'])
pprint.pprint(send_command)
```

Ctrl+O ➜ Enter
Ctrl+X

student@ubuntu22-04~$ `python3 script4_napalm.py`

```
{'display version': 'Huawei Versatile Routing Platform Software\n'
                    'VRP (R) software, Version 5.130 (AR2200 V200R003C00)\n'
                    'Copyright (C) 2011-2012 HUAWEI TECH CO., LTD\n'
                    'Huawei AR2220 Router uptime is 0 week, 0 day, 0 hour, 58 '
                    'minutes\n'
                    'BKP 0 version information: \n'
                    '1. PCB      Version  : AR01BAK2A VER.NC\n'
                    '2. If Supporting PoE : No\n'
                    '3. Board    Type     : AR2220\n'
                    '4. MPU Slot Quantity : 1\n'
                    '5. LPU Slot Quantity : 6\n'
                    '\n'
                    'MPU 0(Master) : uptime is 0 week, 0 day, 0 hour, 58 '
                    'minutes\n'
                    'MPU version information : \n'
                    '1. PCB      Version  : AR01SRU2A VER.A\n'
                    '2. MAB      Version  : 0\n'
                    '3. Board    Type     : AR2220\n'
                    '4. BootROM  Version  : 0\n'}
```

student@ubuntu22-04~$ `ls -l`

```
-rw-rw-r-- 1 student student  93 Apr 23 06:05 ospf_config
-rw-rw-r-- 1 student student 449 Apr 23 06:06 script5_napalm.py
```

student@ubuntu22-04~$ `nano ospf_config`

```
ospf 1 router-id 1.1.1.1
area 0
interface g0/0/1
ip address 10.0.1.1 30
ospf enable 1 area 0
```

Ctrl+O ➜ Enter
Ctrl+X

student@ubuntu22-04~$ `nano script5_napalm.py`

```
from napalm import get_network_driver
import pprint

driver = get_network_driver('huawei_vrp')
device = driver(hostname='172.16.128.1', username='user2', password='user@123')
device.open()

command = ['display current-configuration interface g0/0/1']
get_interface = device.cli(command)
pprint.pprint(get_interface)

device.load_merge_candidate('ospf_config')
device.commit_config()

get_interface = device.cli(command)
pprint.pprint(get_interface)
```

Ctrl+O ➜ Enter
Ctrl+X